

Integration, ETL, and Backup Operations



Estimated time: 60 minutes

Welcome to the lab on Integration, ETL, and Backup Operations!

Learning objectives

In this lab, you will be able to:

- Create ETL and data analytics script using Bash
- Write a Bash script to backup MySQL database server

About SN Labs and Cloud IDE

Skills Network (SN) Labs offer a fully functional environment - Cloud IDE - for course and project-related labs. It runs on Theia, an open-source integrated development environment (IDE) compatible with desktop and cloud platforms.

With this Cloud IDE, you can access Theia and MySQL, both running in a Docker container. It has all the essential tools needed to complete your lab efficiently.

Templates to be used

To complete this lab, you will require the following data files and templates.

- Data files:
 - [sales_file1.csv](#)
 - [sales_file2.csv](#)
 - [product_launch_data](#)
- Templates for Bash script:
 - [template1_bash_script.sh](#)
 - [template2_bash_script.sh](#)

Note: To download data files and templates for the Bash script, right-click on the link and select **Open link in new tab**.

Consideration to work in the lab environment

- This lab environment does not save your session data. Each time you connect, a new environment is created.
- To prevent data loss, plan to complete the lab in a single session.

Introduction

Data integration is a framework for bringing data together from various sources. It provides a unified, consistent view of data and enables organizations to combine data and store them in various formats, databases, and systems into one cohesive structure. Data integration system helps organizations in data analysis, reporting, and making data-driven decisions.

In this lab, you'll focus on how to integrate data systems in FutureMart. You will create data transformation and data validation scripts using a Linux shell. You'll process CSV files and prepare a data pipeline that feeds the fraud detection model. The lab also includes a MySQL backup automation task using bash.

Scenario

FutureMart, a merged entity of TrendyThreads and FashionMart, aims to incorporate a modern, unified enterprise architecture that flawlessly integrates with the online and offline market. Achieving this vision requires planned strategies, robust data migration and data system

integration, and strong data governance.

Until now, you've created a data governance architecture by accessing the existing data architecture of TrendyThreads and FashionMart, designed a typical data architecture for FutureMart, and created a robust data migration plan between TrendyThreads' digital environment and FashionMart's.

In this lab, you aim for effortless data integration, creating ETL and data analytics Bash scripts and developing data pipelines.

This lab consists of four exercises. Complete each exercise one by one and save your responses in the given files. This lab will help you complete tasks for the capstone project.

Let's begin.

Exercise 1: Create ETL script using Bash

FutureMart wants to analyze the data of the product returns. The product return data is available across two files.

1. **sales-file1.csv** contains the columns Sale_ID, Price, Date_Returned of the product.
2. **sales-file2.csv** contains the columns Sale_ID, Product_Name, Category of the product.

In this exercise you will perform the tasks listed below and create a new csv file that the data analytics team will eventually leverage to understand the customer and product return patterns.

1. Extract data from two separate CSV files.
2. Clean the extracted data by removing duplicate rows from each data file.
3. Combine the cleaned data into a single CSV file

To complete this exercise, run the below command to download the data files and a template for the Bash script.

- Run the command below to download **sales-file1.csv**

```
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/WVMRUQinFBvuUHJcF9gi7A/sales-file1.csv
```

- Run the command below to download **sales-file2.csv**

```
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/Q_z8vW_3d0sJ-hTcM4tJcw/sales-file2.csv
```

- Run the command below to download **template1-bash script**.

```
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IxDvec2K12xj9GP10H_t2w/template%201-bash%20script.sh
```

In your favorite text editor, open the template1-bash script.sh and complete the steps given below to create an ETL script using Bash.

Note: Don't delete comments in the Bash script; they will help to evaluate your code.

Step 1: Extract columns

In the **template1-bash script.sh**, under Step 1, write a single line of code to extract columns (Product_Name,Category) from **sales-file2.csv** and save the output in the new file **sales_extracted2.csv**.

Step 2: Transform: Remove duplicates from sales_file1.csv

In the **template1-bash script.sh**, under Step 2, write a single line of code to remove duplicates from **sales-file1.csv** and save it as a new file **sales_uniq1.csv**.

Step 3: Transform: Remove duplicates from sales_extracted2.csv

In the **template1-bash script.sh**, under Step 3, write a single line of code to remove duplicates from **sales_extracted2.csv** and save it as a new file to **sales_uniq2.csv** file.

Step 4: Load: Combine the two files sales_uniq1.csv and sales_uniq2.csv

In the **template1-bash script**, under Step 4, write a single line of code to combine the two files **sales_uniq1.csv** and **sales_uniq2.csv**. The output of this combination should have columns including, Sales_ID,Price,Date_Returned,Product_Name,Category. Ensure that a comma is used as a delimiter.

Step 5: Save the file

Save the completed ETL script using Bash with a file name **exercise_1_etl.sh** on your computer and keep it handy for the final project.

Exercise 2: Create a data analytics script using Bash

In this exercise, you will create a Bash script that will perform basic data analytics for the products that are launched in the past. You'll first extract data from specific columns, remove the header, filter data and store it, and lastly count the number of products for FutureMart.

To complete this exercise, run the below command to download the data files and a template.

- Run the command below to download product-launch-data.csv

```
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/-CZt8xDn6lFewD4SFTABng/product-launch-data.csv
```

- Run the command below to download template2-bash script.sh

```
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/E704ce-8sfhb1Vah5MVYXg/templete%202-bash%20script.sh
```

In your favorite text editor, open **template2-bash script.sh** file and complete the given steps to create a data analytics script using Bash.

Note: Don't delete comments in the Bash script; they will help to evaluate your code.

Step 1: Remove the header and extract the columns

Open `template2-Bash script.sh` in your favorite text editor.

In the **template2-bash script.sh**, under Step 1, write a single line of code for extracting the column `Product_Name` and `Category` from **product-launch-data.csv**.

Next, write a single line code to remove the header from the CSV file, and save the output as a new file, **extracted_data.csv**.

Step 2: Filter apparel products

In the template Bash script, under Step 2, write a single line of code to select all rows that belong to the 'Apparel' category in **extracted_data.csv** and save the output to **filtered_data.csv**.

Step 3: Sort the filtered product data

In the **filtered_data.csv** file, you can see only one data column. To sort all the data from the extracted data file, in **template2-bash script.sh**, under Step 3, write a single line of code to sort all the data. Save this output as a new file, **sorted_data.csv**.

Step 4: Count the number of products

To count the total number of products, in the **template2-bash script.sh**, under Step 4, line number 20, write a single line of code to count how many times each product occurs and sort the data in descending order of count. Save the output as a new file, **final_data.csv**.

For example:

When you run the command `cat final_data.csv`. You should see an output like this.

```
23 T-Shirts
5 Trousers
14 Jackets
```

Note: The above output is only for understanding purpose. Your output will vary based on the input file.

Step 5: Save the file

Save the completed data analytics script using Bash with a file name **exercise_2_data_analytics.sh** on your computer and keep it handy for the final project.

Exercise 3 (OPTIONAL): Optimize data analytics Bash script

Note: This lab is optional and does not contribute to your final project score, but we strongly encourage you to give it a try.

In **Exercise 2: Create a data analytics script using Bash**, you performed four steps and saved outputs in separate files.

In this exercise, you need to replace all four lines of code that you have written in **Exercise 2** and write a single line of code to optimize the data analytics Bash script.

Step 1: Optimize data analytics Bash script

In this step, review the code from the file **exercise_2_data_analytics.sh**, for lines 8, 12, 16, and 20. Combine codes from all four steps from **Exercise 2** and write a single line of code by replacing intermediate file names with Linux pipes.

Save the completed optimized bash script as **exercise_3_optimize_data_analytics.sh** on your computer.

Exercise 4: Write a bash script to backup the MySQL database server

In this exercise, you'll write a Bash script to back up the database on the MySQL server.

Step 1: Write a Bash script

Your Bash script should include the following variables to backup a database on MySQL server.

- `DBUSERNAME`
- `DBPASSWORD`

- HOST
- DB_NAME
- BACKUP_DIRECTORY

Upon running the Bash script, it should take the backup for the web products database and save them as a file **webproducts_backup_<current_date>.sql**. You should store this file in the backup directory.

For example: If the backup database Bash script ran on 8-Apr-2025, you should name the file **webproducts_backup_2025-04-08.sql**.

Note: You can choose any directory of your choice as a backup directory.

Step 2: Save the file

Save the backup script as **exercise_4_mysql_backup.sh** on your local computer and keep it handy for the final project.

Exercise 5: Design a data warehouse schema optimized for analytics

FutureMart has a website activity monitoring system that tracks user sessions, actions performed, and page visits and records them into an RDBMS. Download the file [sql_schema_for_website_monitoring_tool.txt](#) and review the detailed schema.

Note: To download data files and templates for the Bash script, right-click on the link and select Open link in new tab.

Let's look at the objectives for the data warehouse schema listed below.

- The data warehouse should be based on the start schema style.
- The schema should be optimized for daily analytical reporting.
- Your solution should have all the fact and dimension tables.

In this exercise, you will design a data warehouse schema to store the daily aggregates of the website monitoring system by following steps below.

Step 1: Design the schema

Download the file [template_sql_schema_for_website_monitoring_tool.txt](#) and fill all the sections.

Note: To download data files and templates for the Bash script, right-click on the link and select Open link in new tab.

Step 2: Save the file

Save the completed file as **solution_sql_schema_for_website_monitoring_tool.txt** on your local computer and keep it handy for the final project.

Summary

Congratulations on completing the Lab: Integrate Data Systems for FutureMart! In this lab, you've created Bash scripts for ETL and data analytics. You've also prepared a data pipeline that feeds the fraud detection model and included a MySQL backup automation task using Bash.

Author(s)

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